

MPLADS-SATARK (सतर्क) — Comprehensive Feature Guide

Simple Language Explanations, Real Examples, Analogies, MoSPI Data Findings & Workflows

Target Hackathon: SIH 2026 (Problem Statement 26102)	Scope: All 7 Sentinels + 6 Core Product Architecture Features
Evaluated Dataset: 176,925 Real MoSPI eSAKSHI Records	SHA-256 Hash: 9a5c8df1b038c3527a92bfde6371cfb9b2c3a51f89381e4b37d451296c738e4a

SECTION 1: WHAT IS SATARK & WHAT DOES IT SOLVE?

OVERVIEW

SIMPLE EXPLANATION

Every year, thousands of crores of public tax money are allocated under government schemes like MPLADS for public development (building rural roads, school halls, drinking water pipelines, dispensaries). Currently, government audits happen **after** the money is already spent (post-facto audit). By then, if money was misused, overbilled, or claimed for a fake work, it is too late to recover.

SATARK (सतर्क) is a **preventive AI governance system**. It acts like an intelligent health check or antivirus for public project proposals. Before a District Magistrate approves a sanction or releases a payment tranche, SATARK scans the proposal through **7 distinct AI sentinels**, predicts delay risks, checks contractor cartels, maps GIS coordinates, and warns officers in **under 15 milliseconds**.

💡 Simple Analogy for Judges:

Traditional audit is like doing an autopsy after a patient dies. SATARK is like a live health monitor that warns the doctor *before* a heart attack happens so they can save the patient!

SECTION 2: THE 7 CORE DETECTION SENTINELS (THE FORENSIC ENGINES)

1. **VIDHI-KAVACH (Statutory Compliance & March Rush Shield)**

SENTINEL 1

SIMPLE EXPLANATION

VIDHI-KAVACH enforces statutory legal rules. It scans work titles to ensure public funds are not spent on prohibited items listed in **MPLADS Guidelines 2023 Annexure-I Negative List** (such as private places of worship, commercial trusts, or private properties). It also detects **"March Rush"**—a bad habit where officers sanction projects blindly in the final 10 days of March just to dump remaining budget before the fiscal year ends (violating General Financial Rule 62).

 Real Example:

If a proposal asks for ₹5 Lakhs for "Boundary wall construction around XYZ Mandir/Masjid", VIDHI-KAVACH detects keywords like *Mandir* or *Masjid* (using its Indic Lexicon Transliteration engine) and immediately flags a statutory violation penalty of +50 points.

MOSPI DATASET FINDING

Caught **23,722 statutory violations** across India (including 20,553 Negative List breaches and 3,604 March Rush surges).

2. **PUNAR-DRISHTI (NLP Twin Work & Duplicate Sentry)**

SENTINEL 2

SIMPLE EXPLANATION

PUNAR-DRISHTI prevents double-billing on the same physical asset. Contractors or local bodies sometimes re-submit the same work proposal under slightly different names in the same district to claim money twice. PUNAR-DRISHTI uses **TF-IDF text tokenization and Cosine Similarity** to compare new titles against all existing works in that constituency.

 Real Example:

Work 1: "Construction of PCC Road from Main Road to Ward 4" (Sanctioned 2023)
Work 2: "PCC Road Work near Ward 4 connecting Main Road" (Proposed 2024)
PUNAR-DRISHTI detects a **92% Near-Clone match** and alerts the officer to inspect if this is the exact same road!

MOSPI DATASET FINDING

Caught **10,204 duplicate claims** across India (including 6,965 100% exact title clones).

3. **ARTHA-DARPAN (Cost Benchmark & Overpricing Sentry)**

SENTINEL 3

SIMPLE EXPLANATION

ARTHA-DARPAN stops price-padding and budget inflation. It compares project estimates against the official **Central Public Works Department (CPWD) Delhi Schedule of Rates (DSR 2023-24)** adjusted with state location multipliers and regional peer-group medians. If a standard 50-meter rural road costs ₹4 Lakhs on average in a district, but someone proposes ₹18 Lakhs for it, ARTHA-DARPAN flags it as Critical Price Inflation.

 Real Example:

Proposal: ₹19,50,000 for a Small Community Shed.
CPWD Peer Median for Community Sheds in UP: ₹5,20,000.
ARTHA-DARPAN flags **+275% Critical Inflation** with an estimated excess price risk of ₹14.3 Lakhs!

MOSPI DATASET FINDING

Caught **69,170 cost anomalies** representing **₹3,639.9 Crore** in cumulative excess pricing risk.

4. **CHAKRA-VYUH (Contractor Cartel & Vendor Nexus Sentry)**

SENTINEL 4

SIMPLE EXPLANATION

CHAKRA-VYUH breaks contractor cartels and vendor monopolies. It parses thousands of treasury payment vouchers and connects MPs, Implementing Agencies, and Contractors into an **Interactive Network Graph**. It calculates the **Herfindahl-Hirschman Index (HHI)** to see if open competition is dying in a district because 1 or 2 favorite contractors get 80% to 95% of all contracts.

 Simple Analogy:

If a school canteen holds a "fair tender" for lunch suppliers, but 95 out of 100 contracts always go to the principal's nephew's firm, CHAKRA-VYUH catches the hidden nexus!

MOSPI DATASET FINDING

Caught **47,302 cartel risks** across 27,234 registered contractors and 701 Parliamentary Constituencies.

5. **BHU-DRISHTI (Geospatial & Ghost Asset Radar)**

SENTINEL 5

SIMPLE EXPLANATION

BHU-DRISHTI uses satellite remote sensing grids (ISRO Bhuvan / Leaflet GIS) to locate physical assets. It identifies "**Ghost Assets**"—works where payment vouchers show 100% money released, but no verified GPS geotag coordinates exist on the ground, or where 5 projects are suspiciously claimed at the exact same physical lat/long coordinates within 250 meters.

 Real Example:

Payment voucher released: ₹12 Lakhs for a rural drinking water pump.
GPS Geotag status: *Missing / Unverified*.
BHU-DRISHTI marks it with a **Red Ghost Asset Marker** on the interactive map for mandatory physical site verification!

MOSPI DATASET FINDING

Caught **3,031 Ghost Assets** and 12,320 hyper-local spatial clusters nationwide.

6. **SANKHYA-SATYA (Forensic Digit & Threshold Evasion Sentry)**

SENTINEL
6

SIMPLE EXPLANATION

SANKHYA-SATYA uses forensic mathematical analysis. First, it applies **Newcomb-Benford's Law** (a natural rule that first digits 1, 2, 3 appear far more often than 8 or 9 in genuine financial numbers). Second, it detects "**Tender Splitting**" (**Smurfing**)—where a large ₹15 Lakh project is split into three ₹4,95,000 estimates pegged just ₹5,000 below the mandatory ₹5 Lakh e-tendering ceiling to bypass open tenders.

 Real Example:

Sanction cost: ₹4,98,000.
Why suspicious? It is artificially pegged ₹2,000 below the ₹5,00,000 threshold to avoid publishing an open e-tender. SANKHYA-SATYA flags a **Smurfing Threshold Evasion signal**.

MOSPI DATASET FINDING

Caught **7,710 threshold tender splits** and 54,820 unnatural round-number estimates.

7. 🌲 **VIBHED-NETRA (Multidimensional ML Outlier Sentry)**

SENTINEL 7

SIMPLE EXPLANATION

VIBHED-NETRA uses advanced Machine Learning. It runs a **12-Dimensional Isolation Forest model** across 12 project factors simultaneously (cost, physical progress, execution speed, vendor share, location, category). While a project might look normal on any single parameter, VIBHED-NETRA finds complex multi-factor anomalies that human eyes cannot spot.

MOSPI DATASET FINDING

Caught **37,391 multi-dimensional ML anomalies** (including 7,780 critical outliers).

SECTION 3: THE 6 CORE PRODUCT ARCHITECTURE FEATURES

8. 🎯 **SATARK-DRISHTI (Unified Risk Scorer & Predictive Sentry)**

CORE
FEATURE

SIMPLE EXPLANATION

SATARK-DRISHTI is the central brain of the platform. Instead of confusing the officer with 7 separate scores, it fuses all sentinel outputs into one easy-to-understand **Composite Risk Score (0 to 100)**. It categorizes projects into *Critical Risk (75-100)*, *High Risk (50-74)*, or *Moderate Risk (25-49)*, and predicts expected completion delay duration in months.

MOSPI DATASET FINDING

Identified **10,773 Critical Risk works** and **16,348 High Risk works** across India requiring priority field inspection.

9. 🤖 **SATARK-SAMVAAD (GenAI Investigation Copilot)**

CORE FEATURE

SIMPLE EXPLANATION

SATARK-SAMVAAD is an AI search copilot for government officers. Instead of opening huge Excel sheets with 1.7 Lakh rows, an officer types plain English/Hindi questions. SATARK-SAMVAAD parses the query, searches the 176,925 MoSPI records in real time, and returns matching projects, financial exposure, and risk cards.

💡 **Example Queries Supported:**

- * "Show me the top 10 highest-risk projects"
- * "Show high-risk projects in Bihar where estimate is below 5 lakh"
- * "Which contractors are connected to the most flagged projects?"
- * "Find projects with March Rush violations"

10. **PRASHNA-KAVACH (Explainable AI & Evidence Shield)**

CORE FEATURE

SIMPLE EXPLANATION

PRASHNA-KAVACH answers the officer's single most important question: **"WHY WAS THIS FLAGGED?"** Every high-risk project has a prominent red " **WHY WAS THIS FLAGGED?"** button. Clicking it opens a clean window showing the exact legal rules broken, CPWD overpricing calculations, contractor monopoly percentages, and SHAP technical feature weights, complete with direct **"VIEW EVIDENCE"** links.

11. **SATARK-SIMULATION (Live Proposal Risk Sandbox)**

CORE FEATURE

SIMPLE EXPLANATION

SATARK-SIMULATION is a real-time testing sandbox for District Magistrates. Before an officer approves a proposal, they enter parameters (Title, Cost, Date, District, Vendor, Geotag). SATARK runs all 7 sentinels in ****under 15 milliseconds**** and displays the composite score. If the officer changes the cost from ₹4.98L to ₹5.50L, the score updates instantly, letting them fix violations before spending money.

12. **BHAVISHYA-REKHA (Expenditure Trend & March Rush Forecast)**

CORE
FEATURE

SIMPLE EXPLANATION

BHAVISHYA-REKHA satisfies the problem statement's explicit requirement for **trend analysis**. It charts annual expenditure velocity, exposes the ****March Rush surge**** (showing that 24.8% of annual budgets are sanctioned in the final 10 days of March with a 3.65x higher anomaly risk ratio), and uses linear regression to project forward-looking risk trajectories for upcoming quarters.

13. **SATARK-KARYA (Automated Form GFR-19A Vigilance Case Builder)**

CORE
FEATURE

SIMPLE EXPLANATION

SATARK-KARYA turns AI flags into official legal action. With one click (****"GENERATE CASE FILE"****), it automatically compiles an official, printable ****Form GFR-19A Inspection Memorandum****. It includes project metadata, itemized sentinel violations, a mandatory 5-point field engineer inspection checklist, signature sign-off blocks, and a cryptographic ****SHA-256 dataset hash**** to prove data integrity in court or audit.

SECTION 4: MASTER FEATURE COMPARISON TABLE

SUMMARY

Feature Name	Type	Main Algorithm / Technology	Simple 1-Line Summary
VIDHI-KAVACH	Sentinel 1	Indic Lexicon & Rule Matrix	Blocks prohibited works (temples/trusts) & March Rush budget dumping.
PUNAR-DRISHTI	Sentinel 2	TF-IDF & Cosine Similarity	Catches duplicate twin works submitted under rephrased titles.
ARTHA-DARPAN	Sentinel 3	CPWD DSR Rates & Peer Medians	Catches overpriced, price-padded project estimates.
CHAKRA-VYUH	Sentinel 4	HHI Graph Theory & Ego-Graphs	Exposes contractor cartels and vendor monopolies in constituencies.
BHU-DRISHTI	Sentinel 5	GIS Remote Sensing & Geotags	Locates Ghost Assets disbursed without physical geotag verification.
SANKHYA-SATYA	Sentinel 6	Benford's Law & Smurfing Rules	Catches tender-splitting evasion pegged just below ₹5L ceilings.
VIBHED-NETRA	Sentinel 7	12D Isolation Forest ML	Spots complex multi-parameter financial & execution outliers.
SATARK-DRISHTI	Core Architecture	Composite Scorer (0-100)	Combines all 7 sentinels into one risk score & predicts delay months.
SATARK-SAMVAAD	Core Architecture	GenAI Natural Language Copilot	Lets officers ask plain questions over 176,925 MoSPI records.
PRASHNA-KAVACH	Core Architecture	SHAP Explainable AI Drawer	Answers "WHY WAS THIS FLAGGED?" with clear evidence attribution.
SATARK-SIMULATION	Core Architecture	Sub-15ms Real-Time Sandbox	Tests proposal changes live before signing sanction approvals.
BHAVISHYA-REKHA	Core Architecture	Linear Regression Forecasting	Highlights March Rush spikes & projects future quarterly risk trends.
SATARK-KARYA	Core Architecture	Form GFR-19A Memo Builder	Generates printable legal vigilance notices with SHA-256 hash.